

Monolix Installation Troubleshooting

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Error Message 1

When attempting to install Monolix, the following error was encountered:

The compiler g++ failed with the following errors:

```
g++ --version: Apple clang version 16.0.0 (clang-1600.0.26.4)
Target: arm64-apple-darwin24.1.0
Thread model: posix
InstalledDir: /Library/Developer/CommandLineTools/usr/bin
```

Please make sure that g++ command works properly and try again

Issue 1: Clang vs GCC

Apple's g++ is actually a wrapper around the **Clang compiler** and not GCC, which is required by Monolix.

Solution Steps: Install GCC

To install GCC via Homebrew, run:

```
1 brew install gcc
```

This will install the GCC binaries, including g++-14, in /opt/homebrew/bin.

Temporarily Override g++

To point g++ to the newly installed GCC:

```
1 sudo mv /Library/Developer/CommandLineTools/usr/bin/g++ /Library/
  Developer/CommandLineTools/usr/bin/g++_backup
2 sudo ln -s /opt/homebrew/bin/g++-14 /Library/Developer/
  CommandLineTools/usr/bin/g++
```

Verify that the correct g++ is now in use:

```
1 /Library/Developer/CommandLineTools/usr/bin/g++ --version
```

Expected output:

```
g++ (Homebrew GCC 14.x.x) ...
```

Restore Clang After Installation

After installing Monolix, restore Clang's g++:

```
1 sudo rm /Library/Developer/CommandLineTools/usr/bin/g++
2 sudo mv /Library/Developer/CommandLineTools/usr/bin/g++_backup /
   Library/Developer/CommandLineTools/usr/bin/g++
```

Running the Installer in Rosetta 2

If Monolix is not fully ARM-compatible, use Rosetta 2 for x86 emulation.

Install Rosetta 2

```
1 softwareupdate --install-rosetta
```

Open a Rosetta Terminal

1. Go to **Applications** > **Utilities** and locate the Terminal app.
2. Right-click on Terminal, select **Get Info**, and check **Open using Rosetta**.
3. Open Terminal.

Run the Installer

In the Rosetta-enabled terminal:

```
1 cd /path/to/monolix/installer
2 open MonolixSuite2024R1-macosx-installer.app
```

Error Message 2

During module build:

```
fatal error: 'string' file not found
#include <string>
      ~~~~~~
1 error generated.
```

Solution Steps

1. Ensure you are using GCC instead of Clang:

```
1 export CXX=/opt/homebrew/bin/g++-14
2 export CC=/opt/homebrew/bin/gcc-14
3
```

2. Verify the standard library path. If needed, explicitly add it:

```
1 export CXXFLAGS="-I/opt/homebrew/Cellar/gcc/14.0.0/include  
2 /c++/14.0.0"
```

3. Clean and rebuild:

```
1 cd /Users/rohanpandey/lixoft/monolix/.../build  
2 rm -rf *  
3 /Applications/MonolixSuite2024R1.app/Contents/Resources/  
4 monolixSuite/bin/build
```

4. Run in Rosetta mode if necessary.

Final Notes

If you still face issues, try installing GCC 13 explicitly:

```
1 brew install gcc@13  
2 export CXX=/opt/homebrew/opt/gcc@13/bin/g++-13  
3 export CC=/opt/homebrew/opt/gcc@13/bin/gcc-13
```

Troubleshooting Steps

When encountering the Monolix error:

```
fatal error: 'string' file not found
```

follow the steps below to ensure that the correct C++ headers are accessible.

1. **Install or Update Xcode Command Line Tools:**

Even if you have the full Xcode IDE installed, the Command Line Tools are required. Install them by running:

```
xcode-select --install
```

2. **Check Your Xcode Path:**

Confirm that your system is pointing to the correct Xcode version:

```
sudo xcode-select --switch /Applications/Xcode.app/Contents/Developer
xcode-select -p
```

The `xcode-select -p` command should return a path ending in `Developer`.

3. **Accept Xcode License and Initialize the Toolchain:**

If you recently installed or updated Xcode, you may need to accept its license and run the first launch setup:

```
sudo xcodebuild -license
sudo xcodebuild -runFirstLaunch
```

4. **Check macOS SDK Paths (If Needed):**

Verify the availability of the SDK headers:

```
ls /Library/Developer/CommandLineTools/SDKs
```

If headers are missing, reinstall the Command Line Tools.

5. **Rebuild or Relaunch Monolix:**

After ensuring that your system has proper toolchain access, restart Monolix and attempt to reload the demos.

These steps should resolve the issue where standard library headers like `<string>` are not found.